BSc (Hons) in **Information Technology Specialized in AI**

Faculty of Computing

IT3012 : Intelligent Agents

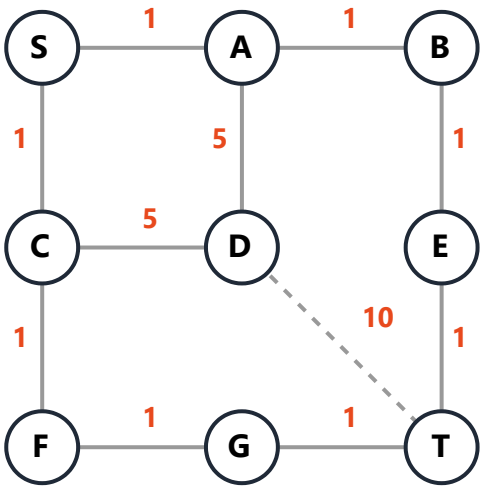
Year 3 · Semester 1 · 2026 Semester 2

Tutorial 04

Select Lab Module: Part 3: Student Work (Treasure Hunt)

Reset Module

The Treasure Map



Heuristic Table: h(n)

Node	S	A	B	C	D	E	F	G	T
h(n)	4	3	2	3	2	1	2	1	0

Treasure Hunt Worksheet

Analyze the map and calculate the final path to the Treasure (T) chosen by each algorithm. Assume ties are broken alphabetically.

1. BFS Path (Shortest Hops)

Format: S-X-Y-T S-A-D **2. DFS Path (Deepest First)**

Format: S-X-Y-T S-A-E **3. A* Search Path (Optimal)**

Format: S-X-Y-T S-A-E **4. Total Cost of A***

Path 4

Verify Answers

All Answers Correct!

- **BFS** fell into the Swamp trap because it blindly searches for fewest hops (3 edges).
- **DFS** luckily found an optimal path due to alphabetical tie-breaking, but it's not guaranteed.

- **A*** perfectly calculated the true costs and bypassed the high-cost Swamp.